

SmartHire - Vision Document

Document Metadata	Value
Group	9
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Revision History

Version	Date	Author/Editor	Summary	Status
1.1	2026-07-10	Nguyễn Gia Quốc Uy and Group 9	Reconciled the project scope, multi-tenant authorization model, feature priorities, measurable quality targets, company-membership workflow, references, and document presentation.	Draft for Team Review

Changes from PA1 Proposal & How this document was developed

This Vision Document reflects several deliberate refinements made since the initial PA1 proposal, based on deeper technical and product discussions within the team.

- **AI Job Description Generator: Removed.**
 - Generating a quality JD requires inputting nearly as much detail as writing it manually, providing negligible automation benefit.
- **AI Resume Enhancement & Gap Analysis: Removed / merged.**
 - Resume rewriting was dropped due to the risk of AI misrepresenting a candidate's actual qualifications.

- Gap-analysis value is instead delivered through the score explanation already required for the scoring engine (Section 5.3.7), rather than as a separate feature.
- **Application status model: Expanded and clarified.**
 - PA1's original 5-state model (Applied → Screened → Interviewing → Offered → Hired/Rejected) has been refined into a 9-state canonical pipeline (Applied, Viewed, Shortlisted, Interviewing, Offered, Hired, Offer Declined, Rejected, Waitlisted) to accurately reflect the recruiter decision points identified during workflow design.
- **Employer registration model: Redesigned.**
 - PA1 assumed a separate employer registration flow.
 - The current design instead uses a single base Candidate account that can request Recruiter/HR Manager permissions via a multi-tenant company membership model, allowing one user to manage recruitment for multiple companies.
- **Authentication token storage: Changed.**
 - PA1 proposed storing JWTs in browser localStorage/sessionStorage.
 - This was revised to HttpOnly, Secure, SameSite cookies to mitigate XSS-based token theft.
- **Job matching: Re-scoped from AI to rule-based.**
 - What PA1 described as "AI-powered job recommendations" is implemented as tag- and location-based Smart Matching, not semantic AI analysis — the AI/LLM capacity is instead concentrated on CV scoring, where it delivers clearer value.

This Vision Document consolidates sections authored by members of Group 9 and reconciles the product scope, terminology, priorities, and architectural assumptions across the project documentation. Authorship and review credit for each section are recorded below. Open items requiring team confirmation are identified explicitly.

1. Introduction

Author of this Part: Ngô Quốc Tuấn

Student ID: 24127581

1.1. Purpose

This document defines the product vision, positioning, target users, high-level capabilities, constraints, and quality objectives for SmartHire, an AI-assisted applicant tracking system

built for small and medium-sized enterprises in Vietnam. It serves as the shared product baseline for requirements, design, implementation, testing, and stakeholder review.

The document is intended for internal use across product, design, development, and quality-assurance functions. It should be consulted during feature discussions, sprint planning, requirements refinement, and stakeholder presentations to maintain alignment between what is being built and why.

1.2. Scope and Out of Scope

In Scope for the Current Project Release

- Candidate account, profile, and CV management.
- Approved-job discovery, filtering, saving, and application submission.
- Company-scoped recruiter access, job-post management, applicant review, and Kanban pipeline management.
- Hybrid candidate scoring using deterministic matching plus AI-assisted semantic analysis and score explanations.
- Candidate application tracking, email notifications, and in-app alerts.
- Employer verification, company membership approval, job moderation, and essential platform administration.
- Backend security logging and audit records required for troubleshooting, moderation, and accountability.

Recruitment analytics and data export are **P1 (Should)** capabilities: they are planned for the current project but may be deferred until all P0 capabilities are stable.

Out of Scope for the Current Project Release

- AI-generated job descriptions.
- AI resume rewriting or resume enhancement.
- Fully automated candidate rejection or hiring decisions.
- Semantic AI job recommendations; current Smart Matching uses rules based on tags, preferences, and location.
- External calendar synchronization, payroll, onboarding, and full human-resource management.

1.3. References

#	Source	Description
1	Geoffrey A. Moore, Crossing the Chasm	Basis for the product-positioning structure: target customer, unmet need, product category, primary benefit, alternatives, and differentiation.
2	Eric Ries, The Lean Startup (2011)	Supports the problem-first and hypothesis-validation approach used in the Problem Statement.
3	Vietnam Ministry of Planning and Investment, Voluntary National Review 2023 on the Implementation of the Sustainable Development Goals	Provides official context on the composition and importance of SMEs in Vietnam.
4	TopCV and ITviec public product pages	Used as representative local job-platform alternatives in the positioning and competitor analysis.
5	National Assembly of Vietnam, Law No. 91/2025/QH15 on Personal Data Protection	Primary legal reference for personal-data processing and protection; effective from 2026-01-01.
6	Government of Vietnam, Decree No. 13/2023/ND-CP on Personal Data Protection	Supporting personal-data protection requirements applicable to the system.

Web references were last accessed on **2026-07-10**.

1.4. Glossary

Term	Definition
AI	Artificial Intelligence. In SmartHire, AI provides advisory semantic CV analysis and explanations; it does not make final hiring decisions.
ATS	Applicant Tracking System; software used to manage job postings, applications, candidates, and hiring workflows.

CV	Curriculum Vitae; a candidate document containing education, experience, skills, and other professional information.
JD	Job Description; the responsibilities, requirements, and employment details associated with a job posting.
LLM	Large Language Model used by the AI service for contextual CV-to-JD analysis.
Multi-tenant	An authorization model in which company data is isolated and access is granted through a company-specific membership.
P0 / Must	Capability required for the current project release to support the end-to-end core workflow.
P1 / Should	Important capability planned after P0 requirements are stable; it may be deferred without breaking the core workflow.
RBAC	Role-Based Access Control; authorization based on platform roles and company-scoped membership roles.
SME	Small or Medium Enterprise. SmartHire targets Vietnamese SMEs with relatively small recruitment teams.

2. Positioning

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This section uses a Problem Statement to define the need and a Product Position Statement to define SmartHire's target market, primary value, alternatives, and differentiation.

2.1. Problem Statement

Field	Details
The problem of	Vietnamese SMEs managing recruitment manually via disconnected tools (email, spreadsheets) without a structured tracking, screening, or evaluation system. This leads to fragmented, duplicated candidate data and loss of institutional knowledge when recruiters leave.

Affects	- Recruiters & HR staff wasting time on repetitive administrative tasks instead of high-value sourcing/engagement. - Hiring managers lacking visibility into candidate pipelines and evaluation contexts. - Job candidates facing poor, delayed updates, which damages the employer's brand. - Business leadership unable to make data-driven decisions due to a lack of structured recruitment analytics.
The impact of which is	- Inefficient hiring: Manual CV screening is slow and prone to bias or errors. - Candidate stagnation: Strong early applicants are missed or forgotten due to the lack of objective scoring. - Low productivity: Recruiters burn out from copy-pasting data, reformatting CVs, and manual email tasks. - Weakened talent brand: Poor candidate communication hurts the company's reputation. - Zero optimization: Without structured historical data, the business cannot track performance or improve its funnel.
A successful solution would be	A lightweight, user-friendly web platform that: - Automates CV screening and scoring. - Provides a visual recruitment pipeline (Kanban board). - Automates candidate status communications. - Adds P1 recruitment analytics and data export after the core workflow is stable. - Connects candidates to jobs via Smart Matching. Success is defined by cutting administrative tasks by 80%, ensuring no candidates are missed, and enabling zero-setup adoption for SMEs.

2.2. Product Position Statement

Field	Details
For	Vietnamese SMEs (10–500 employees) actively hiring but lacking a structured recruitment system, or the budget/IT capacity for complex enterprise HR software.

Who	Struggle with high volumes of unstructured job applications, manual CV screening, lack of candidate-ranking standards, and poor pipeline visibility (relying on scattered emails/spreadsheets).
The SmartHire	An AI-assisted, zero-install ATS and Smart Job Matching web platform, purpose-built for the Vietnamese SME market and candidate job discovery.
That	Provides recruiters with a visual Kanban pipeline, auto-scored CVs, and automated status emails, while matching job seekers with relevant roles based on profile tags and CV preferences.
Unlike	- Enterprise ATS (Workday, Greenhouse): Too complex, expensive, and oversized for SME recruitment teams. - Generic job boards (TopCV, ITviec): Focus only on candidate sourcing without managing or tracking candidates after application. - Manual methods (Email + Excel): Highly inefficient, prone to human error, and lacks auditable data/analytics.
Our product	Offers a unified workflow: recruiters post roles, auto-score applications, and manage candidates visually; candidates upload CVs and receive instant, tailored job matches. This ecosystem drives network effects: a larger database of matched candidates continuously improves hiring quality and speed.

Core differentiators:

- Smart Job Matching Recommendation (rule-based tags/location)
- Structured ATS pipeline for recruiters
- Automated candidate communication
- Planned recruitment analytics and data export for administrators and recruiters

3. Stakeholder and User Descriptions

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3.1. Stakeholder Summary

Stakeholders are individuals, groups, or organizations that influence the development of the recruitment platform or are affected by its operation. The main **stakeholders** include *the*

project owners, development team, candidates, recruiters, hiring companies, platform administrators, and external service providers.

Stakeholder	Type	Role and interest	Influence
Project Owner / Product Owner	Internal	Defines the product vision, project scope, feature priorities, and expected business value.	High
Development Team	Internal	Designs, implements, tests, deploys, and maintains the system, including the AI-assisted candidate scoring module.	High
Recruiters / HR Managers	External	Create job postings, review applicants, manage recruitment pipelines, and communicate with candidates.	High
Hiring Companies	External	Use the platform to advertise vacancies and recruit suitable candidates efficiently.	High
Candidates / Job Seekers	External	Build professional profiles, search for jobs, submit applications, and monitor application progress.	Medium
Platform Administrators	Internal	Verify recruiters, moderate job postings, manage user accounts, and maintain platform safety.	Medium
Project Supervisor / Academic Reviewer	External	Reviews project quality, requirements, design decisions, implementation, and documentation.	Medium
Third-party Service Providers	External	Provide services such as email delivery, file storage, cloud infrastructure, and AI APIs.	Low to Medium

3.1.1. Stakeholder Analysis

The **Product Owner**, **Development Team**, and **Recruiters** have the greatest influence because they directly determine the system requirements, technical implementation, and operational value of the platform.

Candidates have high interest because the system directly affects their job-search experience, although they normally have less influence over product decisions. Administrators ensure the platform remains trustworthy by preventing fraudulent recruiters, misleading job postings, and abusive behavior.

3.2. User Persona Summary

The platform has three primary user groups: Candidates, Recruiters/HR Managers, and Administrators. Each group has different goals, technical abilities, permissions, and usage patterns.

3.2.1. Candidate

Attribute	Description
User profile	University students, recent graduates, unemployed users, and working professionals seeking new opportunities.
Primary goal	Find suitable jobs and complete applications efficiently.
Main problems	Repetitive application forms, irrelevant search results, unclear application status, and limited feedback from recruiters.
Technical ability	Basic to high familiarity with websites, mobile applications, job portals, and professional networking platforms.
Main activities	Register, build profile, upload CV, search for jobs, save jobs, apply, and track application status.
Expected outcome	Successfully obtain a suitable job and receive clear updates during the recruitment process.

3.2.2. Recruiter / HR Manager

Attribute	Description
User profile	HR employees, recruitment specialists, hiring managers, and authorized company representatives.
Primary goal	Find qualified candidates and manage recruitment activities efficiently.

Main problems	Large numbers of applications, manual CV screening, scattered candidate information, and slow communication.
Technical ability	Medium to high; normally familiar with office software, recruitment platforms, CRM systems, or ATS products.
Main activities	Request recruiter verification, create/link company account, create job posts, review applicants, evaluate candidates, update pipeline stages, and export reports.
Expected outcome	Hire suitable candidates while reducing screening time and recruitment workload.

3.2.3. Platform Administrator

Attribute	Description
User profile	Internal personnel responsible for system operation, content moderation, and account management.
Primary goal	Maintain platform quality, security, reliability, and trust.
Main problems	Fraudulent recruiters, misleading job posts, spam, policy violations, and manual moderation workload.
Technical ability	High; familiar with administrative dashboards and system management tools.
Main activities	Verify recruiters, approve job posts, process & export reports, suspend accounts, monitor platform statistics, and review audit logs.
Expected outcome	Maintain a safe recruitment environment with legitimate users and reliable job information.

3.3. User Environment

The recruitment platform is a responsive web application. Its interface and technical design must support different devices, software platforms, and specific usage contexts for each user group.

3.3.1. Environmental Factors Summary

Environment factor	User group	Description
Device	Candidates	Primarily smartphones (70% of search traffic) and laptops. The job-search and application interfaces must be mobile-friendly.
Device	Recruiters	Primarily desktop computers and laptops because recruitment dashboards, applicant tables, and Kanban boards require larger screens for data-dense viewing.
Device	Administrators	Primarily desktop computers for moderation queues, account management, and analytics.
Operating system	All users	Windows, macOS, Android, and iOS.
Browser	All users	Recent versions of Chrome, Edge, Firefox, and Safari.
Network	All users	The system should operate under normal home, university, office, and mobile network (3G/4G/5G) conditions.
Usage location	Candidates	Home, university, workplace, or transit/mobile environments.
Usage location	Recruiters	Company offices or quiet remote-working environments.
Usage frequency	Candidates	Periodic or frequent use during active job searching.
Usage frequency	Recruiters	Daily use during active recruitment campaigns.
Usage frequency	Administrators	Daily or scheduled operational use.
File handling	Candidates	CV upload in PDF or DOCX format, with a maximum size of 5 MB.
Data sensitivity	All users	Personal information, CVs, contact details, company documents, evaluation notes, and application records must be protected.

Notifications	All users	Notifications should be sent to users via email and in-app notifications to keep them updated on the latest information.
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3.3.2. Task Complexity and Team Size

Note: figures in this subsection (team size, cycle duration, time allocation) are team estimates based on general SME hiring norms, not measured data from a specific survey or company.

- **Recruitment Team Size:** In a typical Small and Medium Enterprise (SME) target user environment, a single hiring task involves **2 to 5 people** (1-2 HR/Recruitment specialists sourcing candidates, and 1-3 Hiring Managers/Technical Leads reviewing profiles and conducting interviews). The application must support collaborative candidate evaluation and status updates without data override conflicts.
- **Trend:** As SMEs grow, the number of collaborators increases, requiring role delegation (e.g., OWNER vs. RECRUITER) to manage access rights.

3.3.3. Task Cycle and Activity Time Allocation

- **Recruitment Cycle:** A typical recruitment campaign task cycle lasts from **15 to 45 days** (from job posting creation to final hiring confirmation).
- **Time Spent on Activities:**
 - **Recruiters:** Spend **70%** of their daily recruitment time on manual CV screening and initial evaluation, and **30%** on interview coordination and feedback collection. The platform aims to reduce CV screening time to under 10% using the AI-assisted scoring system.
 - **Candidates:** Spend an average of **5 to 10 minutes** to complete and submit a single job application. The interface must minimize friction to prevent application abandonment.

3.3.4. Unique Environmental Constraints

- **On-the-go Candidates:** Candidates often browse jobs on public transport or under unstable mobile data networks (3G/4G). The frontend must be lightweight, support fast page loading ($\leq 3s$), and implement auto-save forms to prevent data loss on network drops.
- **Data-Dense Recruiter Workspaces:** Recruiters work on multi-tab browsers and high-resolution monitors. The dashboard needs to display high-density tables and clear visual

indicators (such as the Kanban board) to help them quickly scan candidate status amidst office distractions.

3.3.5. Platform Integration and Interoperability

To fit into the existing workflow of SMEs, the application must interact with other standard office tools:

- **Email Systems:** Integration with standard mail protocols to deliver automated application status updates to Gmail and Outlook.
- **Office Productivity Suites:** Capability to export candidate lists and analytics to CSV and Microsoft Excel (.xlsx) for internal company reporting.
- **Calendars:** In future updates, the application should plan for integrations with external calendars (like Google Calendar) to sync interview schedules.

3.4. Summary of Key Stakeholder and User Needs

3.4.1. Key Problems and Desired Solutions

Problem	Reasons for the Problem	Current Workaround (How it is solved now)	Desired Solution (What solution is wanted)	Relative Importance
Recruiters spend too much time screening CVs manually.	CVs are unstructured; comparing skills with job descriptions is subjective and slow.	Reading CVs one by one, using simple Ctrl+F keywords, or outsourcing initial screening.	An AI-powered hybrid scoring algorithm to automatically rank CVs based on skills/experience.	Critical (P0 / Must)
Candidates face lack of transparency in application status.	Recruiters rarely update candidates on progress, leading to anxiety and ghosting.	Keeping manual spreadsheets; sending follow-up emails that go unanswered.	A real-time status tracker tied to a Kanban board and automated notifications at each stage.	Critical (P0 / Must)

High risk of fake recruiters and spam job postings.	Anyone can register and post jobs without verification of business legitimacy.	Reactive reporting after candidates get scammed; manual post-moderation.	Mandatory recruiter business license verification (Tax ID) and Admin approval queues.	Critical (P0 / Must)
Fragmented recruitment coordination within hiring teams.	Recruiters, HR staff, and Hiring Managers communicate through separate chat applications or email.	Sharing Excel sheets, printing CVs, or holding lengthy synchronization meetings.	An interactive Kanban board serving as a shared pipeline for status updates and comments.	Critical (P0 / Must)
Repetitive application process for candidates.	Most job portals redirect to external sites, forcing candidates to re-enter data manually.	Copy-pasting data from resume text into custom application forms repeatedly.	Profile data reuse (1-click apply) and automated parsing of uploaded CVs.	Critical (P0 / Must)

3.4.2. Detailed Stakeholder and User Needs List

The following requirements map the identified needs to user groups. **Must** corresponds to **P0** and is required for the core release; **Should** corresponds to **P1** and is implemented after P0 capabilities are stable.

ID	User or stakeholder	Priority	Need	Description
NEED-01	Candidate	Must	Secure account management	Register, log in, recover passwords, and manage accounts securely.

NEED-02	Candidate	Must	Complete professional profile	Manage personal info, education, work experience, and skills.
NEED-03	Candidate	Must	CV management	Upload, view, replace, and delete CV.
NEED-04	Candidate	Must	Relevant job discovery	Search and filter jobs by keyword, location, salary, experience, and job type.
NEED-05	Candidate	Must	Simple application process	Reuse existing profile/CV info to reduce repetitive data entry.
NEED-06	Candidate	Must	Application transparency	See the real-time status of each submitted application.
NEED-07	Candidate	Must	Timely notifications	Receive updates when recruiters change application status.
NEED-08	Recruiter	Must	Job-post management	Create, save, edit, publish, close, and extend job postings.
NEED-09	Recruiter	Must	Applicant management	View applicants and access their profiles, CVs, and cover letters.
NEED-10	Recruiter	Must	Candidate ranking	Compare candidate qualifications with job requirements and sort applicants.
NEED-11	Recruiter	Must	Recruitment pipeline	Recruiters must be able to move candidates through stages such as Applied, Viewed, Shortlisted, Interviewing, Offered, Hired, Offer Declined, Rejected, and Waitlisted.
NEED-12	Recruiter	Should	Candidate evaluation	Write feedback notes and assign evaluation ratings.

NEED-13	Recruiter	Must	Automated communication	Automatic email and in-app notifications triggered by pipeline status changes.
NEED-14	Administrator	Must	Recruiter verification	Verify company registration documents before allowing job publication.
NEED-15	Administrator	Must	Job moderation	Approve, reject, or request revision of job postings.
NEED-16	Administrator	Must	User management	Search, suspend, and reactivate user accounts.
NEED-17	Administrator	Must	Backend auditability	Record security-sensitive administrative and recruitment actions in system logs.
NEED-18	Product Owner	Should	Business visibility	Dashboard showing platform growth, active postings, and conversion rates.
NEED-19	Development Team	Must	Maintainable architecture	Modular components, consistent REST APIs, and decoupled state management.
NEED-20	All users	Must	Privacy and security	Secure personal data and CV files against unauthorized access.

3.5. Alternative and Competing Solutions

The platform competes with both job-search websites and Applicant Tracking Systems. It may also replace manual recruitment methods, in-house builds, and informal channels currently used by small organizations.

Alternative or Competitor	Category	Main Strengths	Limitations or Opportunity for SmartHire
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LinkedIn	Professional networking and recruitment platform	Established professional networks, company pages, a large talent pool, and strong brand recognition.	Paid recruiter services and sponsored posts may be costly for SMEs, while post-application tracking still requires a separate workflow.
Indeed	Job-search and job-advertising platform	Broad reach and a straightforward job-posting experience.	Primarily supports candidate acquisition; growing application volumes may still require export to email, spreadsheets, or an ATS.
TopCV / VietnamWorks / CareerViet / ITviec	Local job platforms	Familiar Vietnamese recruitment brands, local job coverage, and established candidate audiences.	Applications remain separated by platform, making it difficult for recruiters to maintain a unified cross-source pipeline.
Greenhouse / Lever	Applicant Tracking Systems	Structured pipelines, interview workflows, integrations, and mature recruitment operations.	Pricing, configuration, and operational complexity may be difficult to justify for SMEs with small recruitment teams.
Workday	Enterprise HR and ATS platform	Broad HR capabilities, compliance support, analytics, and enterprise reporting.	Implementation cost, setup effort, and required IT support are generally beyond the needs of the target SME segment.
In-house build	Custom internal solution	Can be tailored to an organization's current processes and data model.	Requires continuing development and maintenance capacity and creates knowledge-continuity risks when key developers leave.

Spreadsheets and email	Manual status quo	Familiar, flexible, and inexpensive to start.	Candidate records become duplicated or outdated, there is no reliable single source of truth, and follow-up remains manual.
Social media groups (Facebook, Zalo groups)	Informal recruitment alternative	Low-cost distribution and access to passive candidates.	Limited identity and job-post verification; screening through comments and direct messages does not scale reliably.
Company career pages	Direct recruitment channel	Full control over employer branding and the candidate-facing experience.	Limited organic reach and no replacement for the internal application-management workflow provided by an ATS.

4. Product Overview

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The SmarHire Platform is a recruitment system designed to automate the hiring lifecycle for Small and Medium Enterprises (SMEs) by integrating job management, candidate tracking, and AI-assisted hiring features into a centralized web-based solution.

The platform utilizes a modern technical stack and a decoupled architecture:

Architecture & Security

- Follows a **Decoupled Client-Server Architecture**.
- Uses **JSON Web Tokens (JWT)** for stateless authentication and session management.
- Implements **Role-Based Access Control (RBAC)** combined with a **Multi-tenant model**, where a base user account acts as a Candidate, but can simultaneously hold Recruiter or HR Manager permissions for one or multiple companies via company membership records.

Frontend Stack

- Built with:
 - **Next.js (React)**
 - **TypeScript**
 - **Tailwind CSS**
 - **Shadcn UI**
- Uses **Zustand** for lightweight state management.
- Integrates **hello-pangea/dnd** to support drag-and-drop functionality in the Kanban board interface.

Backend & Database

- Backend follows a **Layered Architecture** implemented within **Next.js API Routes**.
- Uses a relational database management system:
 - **PostgreSQL**, or
 - **MySQL**
- Provides transactional integrity and reliable data management.

Operating Environments

- Configured as a **responsive web application**.
- Supports:
 - **Desktop View**: Optimized for administrative and management tasks with data-dense interfaces.
 - **Mobile/Tablet View**: Optimized for candidates and users who need access while on the go.

4.1. Product Perspective

SmarterHire is a standalone, AI-assisted recruitment management platform that operates as a responsive web application. It serves as a centralized hub connecting job candidates, recruiters, and system administrators throughout the recruitment lifecycle.

The system uses a client-server architecture in which the frontend communicates with backend services through secure RESTful APIs. SmarterHire integrates with an AI service for semantic CV analysis and human-readable score explanations, an email service for notifications, and a relational database for persistent data storage.

SmartHire replaces traditional recruitment methods that rely on spreadsheets, emails, and manual tracking with an automated and structured recruitment workflow. The platform supports the complete hiring process, from job creation and candidate application submission to screening, interviewing, and final hiring decisions.

The major system components include:

Candidate Portal

- Profile management
- CV upload, parsing, and management
- Job searching and application submission
- Application tracking and access to disclosed score explanations

Recruiter Portal

- Job posting management
- Applicant screening and evaluation
- Kanban-based recruitment pipeline

Administration Portal

- User and recruiter verification
- Job post moderation
- Platform monitoring and analytics
- System audit management

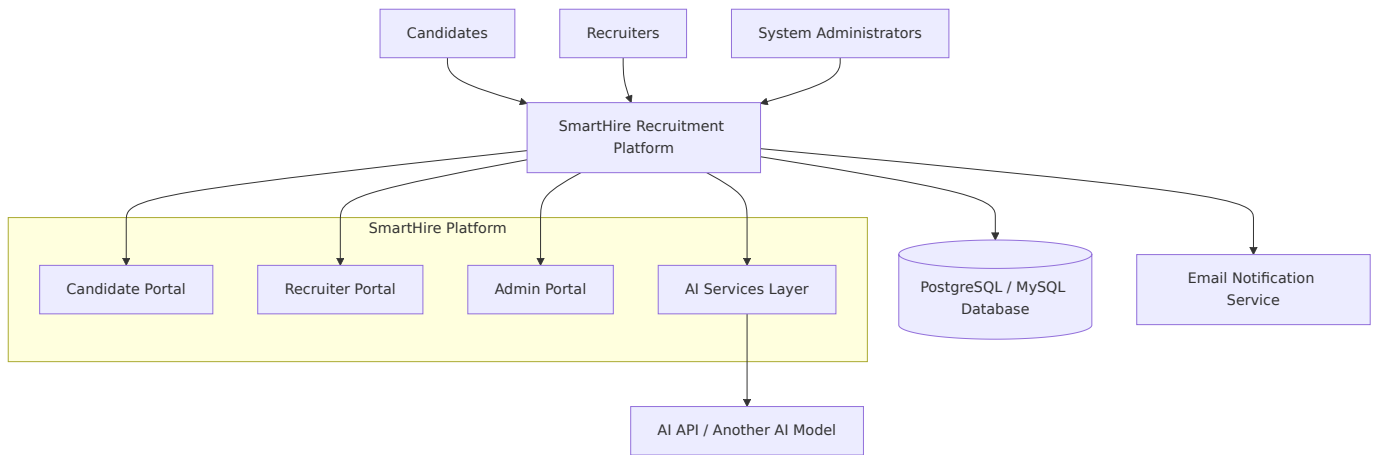
AI Services Layer

- Hybrid candidate scoring (rule-based matching + AI semantic analysis)
- Human-readable score explanations (shared with both recruiter and candidate)

External Systems

- AI API or custom AI model
- Email notification service
- Relational database system (PostgreSQL/MySQL)

The overall product ecosystem can be represented as:



4.2. Assumptions and Dependencies

The successful operation of SmartHire depends on several assumptions and external dependencies.

Assumptions

- Users have access to a stable internet connection to interact with the platform.
- Candidates possess resumes in supported formats (`.pdf` or `.docx`) for upload and processing.
- Recruiters provide accurate job information and valid business verification documents.
- Users access the platform through modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.
- AI-generated recommendations and scoring results are used as decision-support tools rather than fully automated hiring decisions.
- System administrators actively review recruiter registrations and job postings to maintain platform quality, security, and compliance.

Dependencies

AI Service Dependency

- The platform depends on AI API to provide semantic CV scoring and human-readable score explanations.
- AI-powered features may become unavailable or limited if these services experience downtime or API restrictions.

Email Service Dependency

- Password recovery, application status updates, interview invitations, offer letters, and other notifications require a reliable email delivery service.

- Service interruptions may delay communication between recruiters and candidates.

Authentication Dependency

- The platform relies on JSON Web Token (JWT) technology to provide secure authentication, session management, and role-based access control.
- Security mechanisms must remain operational to prevent unauthorized access.

Database Dependency

- SmartHire requires a relational database management system such as PostgreSQL or MySQL to store user accounts, job postings, applications, recruiter verification records, and system logs.
- Database failures may affect system availability and data integrity.

Hosting and Infrastructure Dependency

- The application requires cloud or server infrastructure capable of hosting the frontend, backend services, database, and AI integrations.
- Continuous availability depends on server uptime, network connectivity, and infrastructure maintenance.

Legal and Regulatory Dependency

- SmartHire must comply with applicable Vietnamese personal-data protection requirements, including Law No. 91/2025/QH15 and supporting requirements under Decree No. 13/2023/ND-CP where applicable.
- Future regulatory changes may require modifications to data storage, privacy policies, and user consent mechanisms.

These assumptions and dependencies form the foundation upon which SmartHire's functionality, security, scalability, and user experience are built.

5. Product Features

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5.1. Feature Overview

The table below summarizes the high-level feature groups of the SmartHire system. Each group may include multiple sub-features, which are described in the following section.

Priority scheme:

- **P0 (Must):** required for the current project release and the end-to-end recruitment workflow.
- **P1 (Should):** important but implemented only after P0 capabilities are stable; it may be deferred without breaking the core workflow.

5.2. Detailed Feature List

No	Group Feature	Short Description	Priority
1	Authentication, Authorization & Access Control	Allows users to register and log in securely, supports email verification and password recovery, and enforces platform-level and company-scoped authorization.	P0 (Must)
2	Account Setup & Management	Allows users to update non-critical account information, profile images, and preferences. Password changes remain part of P0 authentication security.	P1 (Should)
3	Candidate Profile Management	Allows candidates to build a professional profile through a form or CV upload, with a CV parser that standardizes data for applications and scoring.	P0 (Must)
4	Job Board & Advanced Search	Allows candidates to search, filter, view, save, share, report, and apply to approved job postings.	P0 (Must)
5	Job Posting Management	Allows company-authorized recruiters to create, preview, edit, and manage the lifecycle of job postings.	P0 (Must)
6	Application Tracking (Candidate Side)	Allows candidates to track saved jobs, submitted applications, scoring progress, and recommended jobs.	P0 (Must)
7	Candidate Screening & Hybrid Scoring System	Combines deterministic skills/experience matching with AI-assisted semantic analysis to score and rank applicants for a job posting.	P0 (Must)

8	Recruitment Pipeline Kanban Board	Provides a drag-and-drop interface for authorized company members to track and update application stages.	P0 (Must)
9	Automated Notifications & In-App Alerts	Sends email and in-app notifications when relevant application or moderation events occur.	P0 (Must)
10	Job Posting Moderation & Quality Assurance	Allows administrators to approve or reject job postings and handle spam or violation reports.	P0 (Must)
11	User Management & Employer Verification	Allows administrators to find user accounts, verify company documents, approve memberships, and handle violations.	P0 (Must)
12	Recruitment Analytics & Data Export	Provides dashboards, recruitment statistics, and CSV/Excel exports for authorized users.	P1 (Should)

5.3. Feature Descriptions

5.3.1. Authentication, Authorization & Access Control

This feature group covers registration, authentication, session security, and authorization. A standard user retains candidate capabilities and may also receive `OWNER`, `HR_MANAGER`, or `RECRUITER` permissions through an approved membership in one or more companies.

Recruitment permissions are therefore scoped to a company rather than stored as a replacement for the user's candidate identity. Platform Administrators hold a separate platform-level role for verification, moderation, account management, and audited support operations. The backend validates both the user's platform role and active company membership before granting access to company data.

5.3.2. Account Setup & Management

This feature allows users to manage contact details, profile images, preferences, and other non-critical account information. Company information is managed separately and only by members with the required company-scoped permission. Password and session-security operations remain part of the P0 authentication feature group.

5.3.3. Candidate Profile Management

This feature allows candidates to create a structured professional profile through a form or by uploading a PDF or DOCX CV. The CV parser extracts profile data for candidate review and reuse in applications and provides normalized text to the matching and scoring service.

5.3.4. Job Board & Advanced Search

This feature provides a public catalogue of approved job postings. Candidates can search and filter by criteria such as salary, experience, location, job type, and relevant tags, then view, save, share, report, or apply to a posting.

5.3.5. Job Posting Management

This feature allows company-authorized recruiters to create, preview, edit, submit, publish, close, and extend job postings according to the approved lifecycle. Structured requirements such as skills, experience, salary range, location, and job type provide input to search, matching, and candidate screening.

5.3.6. Application Tracking (Candidate Side)

This feature provides candidates with a consolidated view of saved jobs, submitted applications, scoring progress, and rule-based job recommendations. The canonical recruitment stages are **Applied**, **Viewed**, **Shortlisted**, **Interviewing**, **Offered**, **Hired**, **Offer Declined**, **Rejected**, and **Waitlisted**. AI-processing progress is displayed separately through `scoring_status` and does not alter the recruitment stage.

5.3.7. Candidate Screening & Hybrid Scoring System

This feature evaluates candidate applications for a specific job posting through a hybrid approach that combines rule-based skills and experience matching with AI-assisted semantic CV analysis. It produces a compatibility score and explanation to support recruiter review, while leaving all progression and hiring decisions under human control.

5.3.8. Recruitment Pipeline Kanban Board

This feature provides authorized company members with a visual representation of applications by recruitment stage. Dragging a candidate card to an allowed stage updates the application transactionally, records the actor and time, and triggers any configured notifications.

5.3.9. Automated Notifications & In-App Alerts

This feature communicates application and moderation events without requiring users to poll the system manually. When an authorized recruiter changes an application stage, the

system records the event and sends the configured email and in-app notification while preventing duplicate delivery.

5.3.10. Job Posting Moderation & Quality Assurance

This feature allows Administrators to review job postings for legitimacy, completeness, spam, and policy violations. A new posting remains unavailable on the public job board until an Administrator approves it; rejection and revision requests include a recorded reason.

5.3.11. User Management & Employer Verification

This feature allows Administrators to search user accounts, perform audited support or enforcement actions, verify company documents, and review company-membership requests. A tax-ID match never grants access automatically: joining an existing company also requires a valid invitation or approval from an existing **OWNER** before final administrative verification.

5.3.12. Recruitment Analytics & Data Export

This P1 feature provides authorized recruiters with company-scoped metrics such as posting views, application counts, stage conversion, and successful hires. Administrators receive platform-level aggregate metrics. Authorized users may export permitted data to CSV or Excel without gaining access to records outside their tenant or platform role.

5.4. Key User Workflows

5.4.1. Hybrid CV Scoring System

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Description

The scoring system uses a hybrid algorithm that combines automatic skill/experience matching (Auto Matching) with AI-based CV analysis (LLM) to evaluate how well a candidate fits the JD.

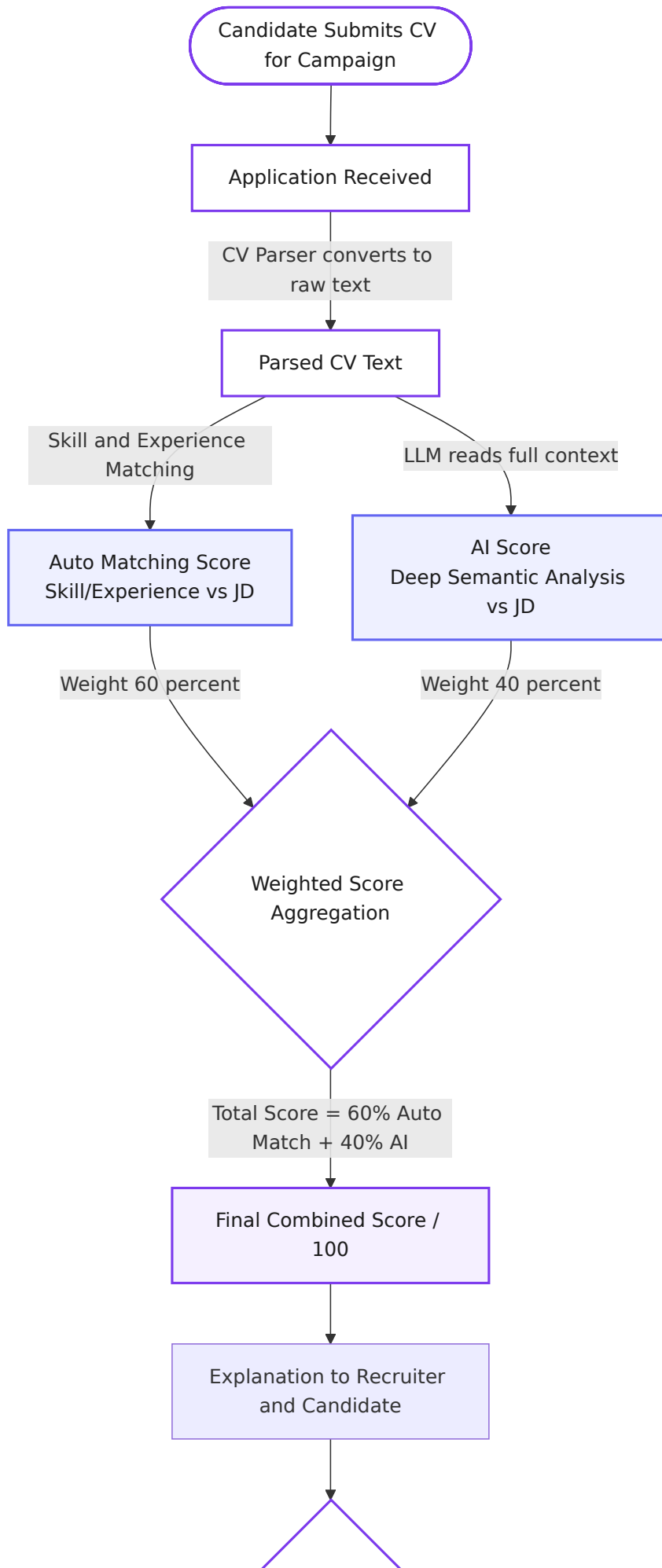
Scoring formula:

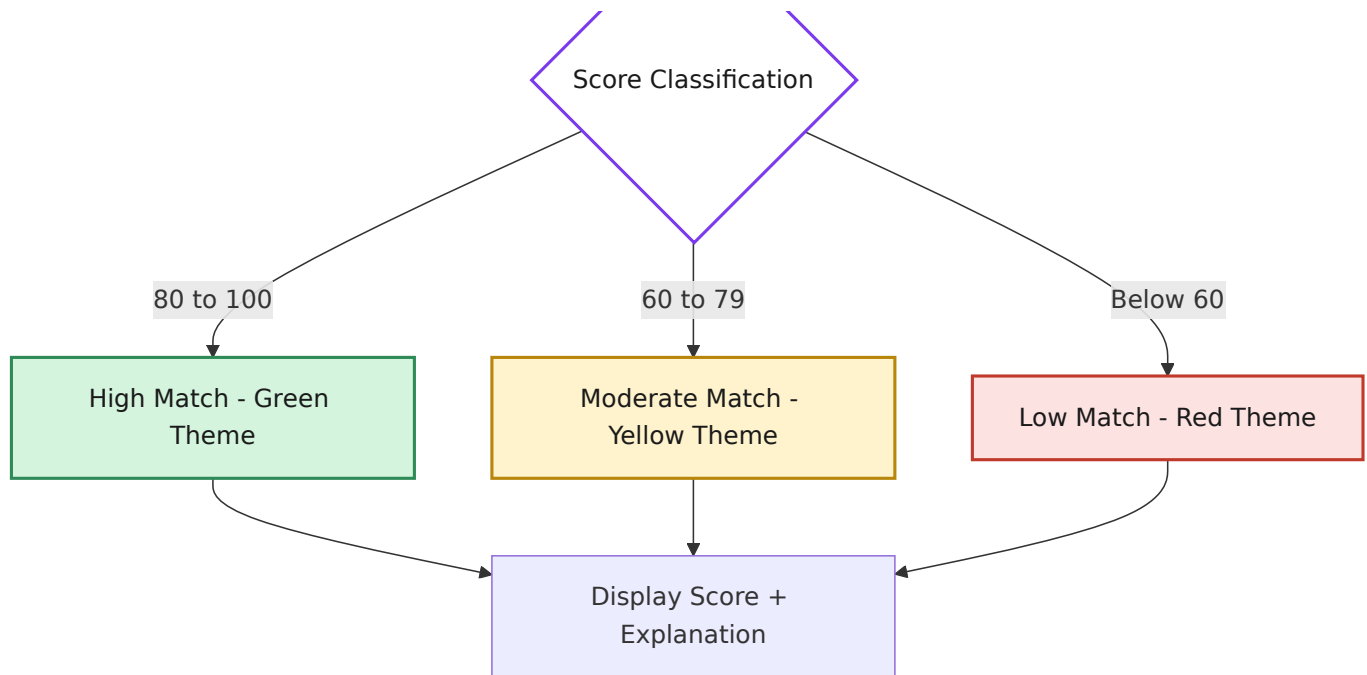
$$\text{Final Score} = 60\% \times \text{Auto Matching Score} + 40\% \times \text{AI Score}$$

Score classification (out of 100):

Score Range	Match Level	Color Theme
80 - 100	High Match	 Green
60 - 79	Moderate Match	 Yellow
0 - 59	Low Match	 Red

Flowchart



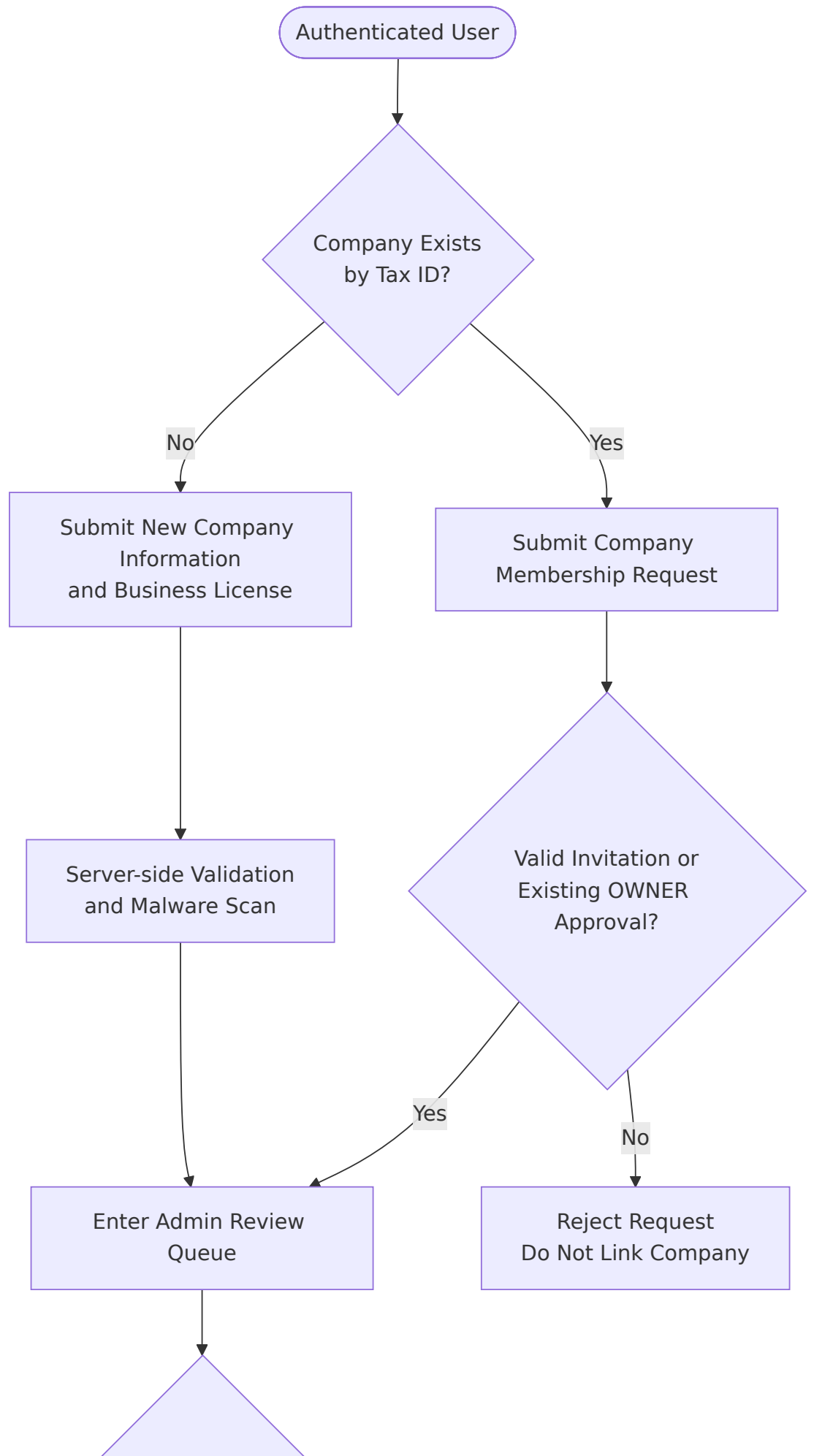


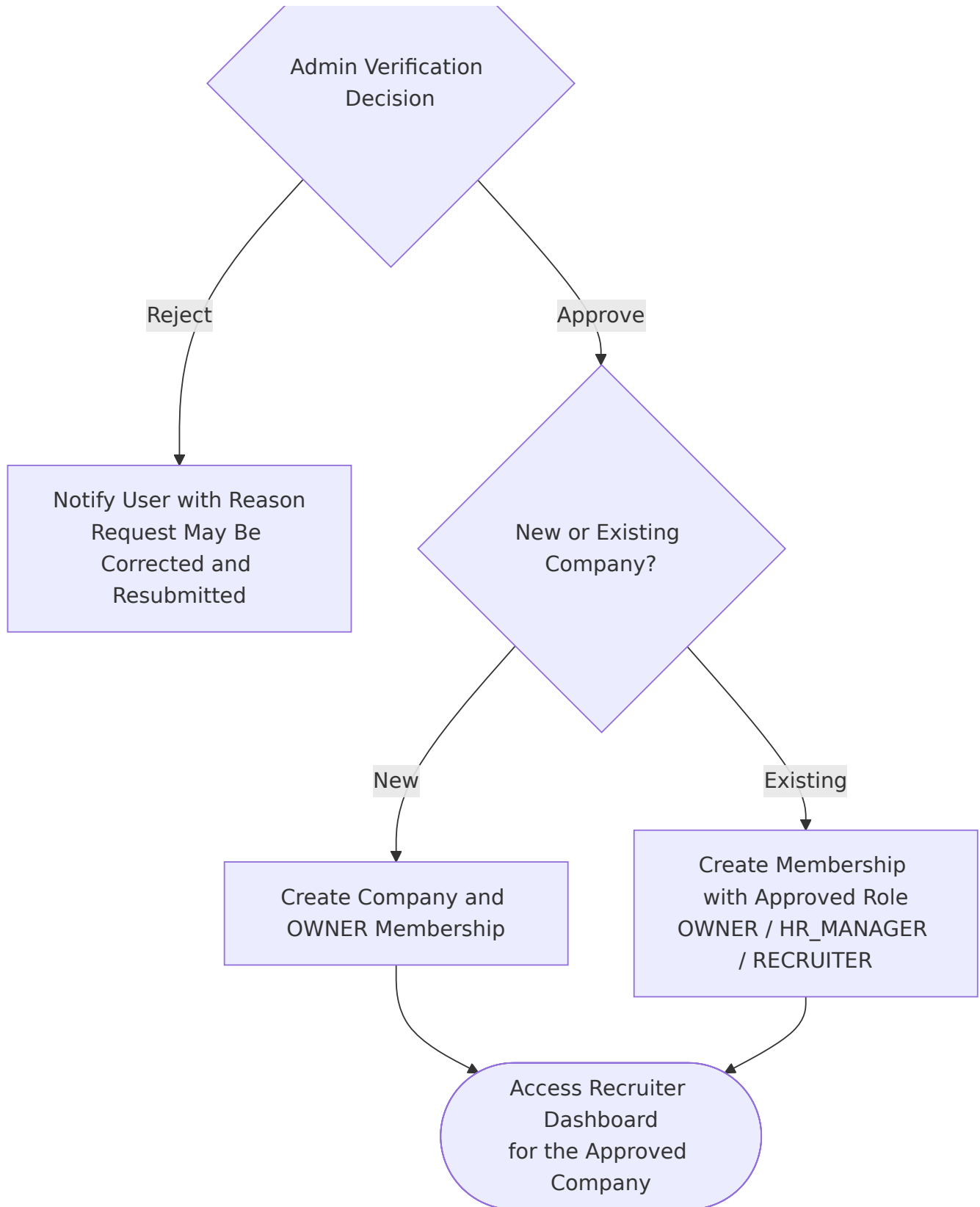
Step-by-Step Explanation

1. **Application Received:** The candidate submits a CV and cover letter for a specific recruitment campaign.
2. **Parsed CV Text:** The CV Parser converts the original CV into normalized raw text.
3. **Auto Matching Score:** An algorithm directly compares the skills/experience in the CV against the JD requirements.
4. **AI Score:** An LLM reads the entire text, understands deep context (handling abbreviations and mixed languages), and compares it against the JD to score each criterion in detail.
5. **Weighted Score Aggregation:** The two scores are combined using a weighted formula: 60% (Auto) / 40% (AI).
6. **Final Combined Score:** The final aggregated score on a scale of 100.
7. **Score Explanation Generation:** The LLM generates a human-readable explanation describing why the candidate received that score, identifying strengths and gaps.
8. **Score Classification:** The score is classified and assigned a color theme (green/yellow/red) based on thresholds for visual display.
9. **Display Score:** The score and explanation are displayed to the recruiter and candidate.

5.4.2. Recruiter Verification & Company Role Assignment

This workflow governs how a standard user receives company-scoped recruitment permissions under the multi-tenant model described in Section 5.3.1. Approval creates a company membership; it does not replace the user's candidate identity. A tax-ID match is used only to identify an existing company and never grants membership automatically.





Authorization safeguards: joining an existing company requires a valid invitation or explicit approval from an existing `OWNER`, followed by Admin verification. New-company requests require validated company information and a malware-scanned business document. All approval, rejection, role-change, and revocation events are written to the backend audit log.

Post-approval lifecycle: a membership may later change when an Administrator revokes access, a user leaves a company, an `OWNER` transfers ownership, or the company is

deactivated. Company deactivation unpublishes its job postings and disables its memberships. The user's candidate profile remains unaffected because only company-scoped permissions change.

Design rationale: recruitment permissions remain outside the user's core identity because one person may recruit for multiple companies simultaneously. Storing a single recruiter role directly on the user record would not represent that relationship or enforce tenant isolation correctly.

6. Non-Functional Requirements

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6.1. Overview

The following non-functional requirements define the quality attributes, operational constraints, and engineering standards of the SmartHire Recruitment Platform. Unlike functional requirements, these requirements describe how well the system performs rather than what it does.

These requirements apply across all major functional modules, including authentication, candidate profile management, AI-powered CV analysis, job posting management, recruitment pipelines, notifications, analytics, and administrative functions.

6.2. Performance Requirements

The platform shall provide responsive interactions under the following acceptance-test baseline unless a test case states otherwise:

- production build deployed in the staging environment;
- 100 concurrent active users;
- seeded data containing at least 10,000 users, 20,000 job postings, and 100,000 applications;
- simulated client network of at least 10 Mbps download, 2 Mbps upload, and 100 ms round-trip latency;
- response-time targets evaluated at the 95th percentile (P95) over a 15-minute test; and
- HTTP error rate below 1%, excluding intentionally rejected validation and authorization requests.

ID	Requirement	Acceptance Target
PERF-01	Public page loading	The primary content becomes usable within 3 seconds under the baseline client network.
PERF-02	Dashboard navigation	P95 ≤ 2 seconds.
PERF-03	Job search and filtering	P95 ≤ 2 seconds.
PERF-04	Login and registration	P95 ≤ 3 seconds, excluding email-delivery time.

PERF-05	Candidate profile update	P95 $\leq$ 2 seconds.
PERF-06	Kanban drag-and-drop update	Visual feedback $\leq$ 500 ms and server persistence P95 $\leq$ 2 seconds.
PERF-07	In-app notification delivery	P95 $\leq$ 5 seconds after the triggering transaction is committed.
PERF-08	CSV/Excel export (P1)	Completed within 10 seconds for up to 10,000 records when the P1 export feature is delivered.
PERF-09	AI semantic scoring	P95 $\leq$ 30 seconds for a supported CV of up to 5 MB; processing remains asynchronous and never blocks the user interface.

If an asynchronous AI request exceeds the target, the system shall keep the scoring status visible, allow other application activities to continue, and notify the user when processing completes or fails.

AI processing shall use a separate `scoring_status` (`Pending` , `Processing` , `Completed` , or `Failed`) rather than adding a temporary value to the canonical recruitment pipeline status.

6.3. Scalability Requirements

The architecture shall meet the following capacity baseline without a major redesign:

The system shall:

- store at least 10,000 registered users, 20,000 job postings, and 100,000 applications.
- support at least 200 simultaneous active users with an HTTP error rate below 1% and P95 response times no more than 20% above the 100-user performance baseline.
- support at least 20 active recruiters within one company without lost updates or cross-company data exposure.
- allow stateless frontend and API instances to be horizontally scaled without changing business logic.
- support database growth through documented indexes, query plans, pagination, and archival procedures.

- support migration to a cloud deployment through environment-based configuration rather than source-code changes.

The AI integration shall be isolated behind a provider interface so that a supported provider or local model can be replaced without changing recruitment-domain logic.

6.4. Availability and Reliability

The SmartHire platform shall provide reliable operation for both recruiters and candidates.

Availability

- Monthly uptime target: **99.5%**, excluding planned maintenance announced at least 24 hours in advance.
- Recovery Time Objective (RTO): **≤ 60 minutes** after a failed deployment or recoverable infrastructure incident.
- Recovery Point Objective (RPO): **≤ 24 hours** for persistent application data.
- Every production deployment shall have a documented rollback procedure that is tested before final release.

Reliability

The system shall:

- prevent data corruption during unexpected failures.
- preserve uploaded CV files.
- ensure recruitment pipeline states remain consistent.
- prevent duplicate applications by enforcing a unique candidate/job-application constraint and idempotent submission handling.
- use transactional database operations for critical updates.

6.5. Security Requirements

Security is one of the highest priorities because the platform stores sensitive personal and business information.

Authentication

The system shall:

- require authenticated login before accessing protected resources.

- issue stateless JSON Web Tokens (JWT).
- securely invalidate user sessions after logout.
- enforce password reset through email verification.
- store authentication tokens using HttpOnly, Secure, SameSite cookies, never in localStorage or sessionStorage, to prevent token theft via client-side script injection (XSS).

Authorization

Authorization shall combine platform-level roles with company-scoped memberships:

- Every standard user account retains access to its own candidate profile and candidate functions.
- Recruitment permissions are granted through an active `CompanyMembership` for a specific company; granting membership does not replace the user's candidate identity.
- A user may hold memberships in multiple companies, and every recruiter request shall include or resolve an active company context.
- `OWNER` , `HR_MANAGER` , and `RECRUITER` membership roles shall receive only the permissions defined for that company role.
- Company members may access job posts, applications, candidate documents, evaluations, and analytics only when those records belong to their active company context.
- `OWNER` may manage company information and company memberships; `HR_MANAGER` and `RECRUITER` permissions shall be limited to their approved recruitment responsibilities.
- Platform `ADMIN` users may perform verification, moderation, account-management, and audited support actions across companies without becoming company members.
- The backend shall verify authentication, platform role, membership status, company scope, and resource ownership on every protected request; frontend route guards are not sufficient authorization controls.

Unauthenticated requests shall return HTTP `401` ; authenticated requests that lack the required platform or company-scoped permission shall return HTTP `403` without exposing data from another tenant.

Password Security

Passwords shall:

- never be stored in plain text.
- be hashed using secure industry-standard hashing algorithms.
- satisfy minimum password complexity requirements.

Data Protection

The system shall:

- encrypt sensitive communication using HTTPS.
- protect against common web vulnerabilities including:
 - SQL Injection
 - Cross-Site Scripting (XSS)
 - Cross-Site Request Forgery (CSRF)
 - Broken Authentication
- validate all user inputs before database processing.
- sanitize uploaded file names.

6.6. File Storage Requirements

The system shall support secure document management.

Resume uploads shall satisfy the following constraints:

Item	Requirement
Supported formats	PDF, DOCX
Maximum file size	5 MB
Duplicate protection	Supported
Malware scanning	Required before a file becomes available to other users or to downstream parsing services
Secure storage	Required

Deleted files shall no longer be publicly accessible.

6.7. AI Service Requirements

The AI components shall function as decision-support tools rather than autonomous decision makers.

The platform shall:

- generate AI recommendations without automatically rejecting candidates.
- allow recruiters to override AI recommendations at any time.
- display generated AI scores together with human-readable explanations.
- continue operating when AI services are temporarily unavailable by falling back to rule-based matching where applicable.

6.8. Usability Requirements

The platform shall meet the following usability acceptance targets:

- at least 90% of representative participants shall complete the primary task for their role without facilitator assistance after no more than 30 minutes of onboarding;
- the primary tasks are candidate application submission, recruiter applicant-stage update, and administrator verification decision;
- usability testing shall include at least five representative participants for each primary user group before final acceptance; and
- the System Usability Scale (SUS) target shall be at least 75.

The interface shall:

- follow consistent navigation patterns.
- complete the primary role workflows with no unnecessary confirmation or repeated data-entry steps.
- provide responsive layouts for desktop, tablet, and mobile devices.
- clearly indicate loading, success, and error states.
- display meaningful validation messages.
- support drag-and-drop interactions for Kanban recruitment management.
- provide searchable and filterable tables.
- provide role-specific guidance sufficient to meet the 30-minute onboarding target.

The primary authentication, application, pipeline, and moderation workflows shall target WCAG 2.1 Level AA, including:

- readable typography

- sufficient color contrast
- keyboard accessibility
- descriptive labels
- responsive layouts

6.9. Compatibility Requirements

At release time, the platform shall operate correctly on the latest two stable major versions of:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

The frontend shall support:

- mobile viewports from 360 px wide;
- tablet viewports from 768 px wide; and
- desktop viewports up to at least 1440 px wide.

Supported resume formats include:

- PDF (.pdf)
- Microsoft Word (.docx)

Supported export formats include:

- CSV
- Microsoft Excel (.xlsx)

6.10. Maintainability Requirements

The software shall be designed for long-term maintenance.

The implementation shall:

- follow modular architecture.
- separate frontend and backend services.
- use layered backend architecture.
- maintain reusable UI components.

- follow TypeScript coding standards.
- use Git for version control.
- support continuous feature expansion.
- pass TypeScript compilation, linting, and automated tests in continuous integration before merge.
- maintain at least 70% automated line coverage for security-sensitive and recruitment-domain service modules.

Business logic, database access, and presentation logic shall remain loosely coupled.

6.11. Database Requirements

The database shall:

- maintain ACID transactional consistency.
- enforce referential integrity.
- automatically generate unique identifiers.
- prevent duplicate critical records.
- support indexing for high-frequency search operations.
- maintain audit information for important administrative actions.

Regular backup procedures shall be supported.

6.12. Notification Requirements

The notification subsystem shall:

- automatically trigger notifications based on recruitment events.
- enqueue interview invitations and status notifications within 5 seconds after the triggering transaction is committed.
- notify candidates when application status changes.
- support configurable email templates.
- prevent duplicate notification delivery.

Temporary email-service failures shall not interrupt other platform operations. Failed deliveries shall be retried at least three times with backoff and recorded for administrative troubleshooting.

6.13. Logging and Monitoring

The requirements below describe mandatory P0 backend audit logging for security, accountability, and debugging. An admin-facing activity-log interface is a separate P1 presentation capability and may be deferred without removing the underlying audit records.

The platform shall maintain system logs for:

- user authentication
- administrator actions
- recruiter moderation actions
- job posting approvals
- account suspension
- AI processing failures
- export activities
- critical system errors

Logs shall support troubleshooting and security auditing.

6.14. Legal and Compliance Requirements

The platform shall comply with applicable legal and ethical standards. The production compliance baseline shall be reviewed before deployment because legal obligations may change during the project lifecycle.

These include:

- Law No. 91/2025/QH15 on Personal Data Protection, effective from 2026-01-01;
- Decree No. 13/2023/ND-CP on Personal Data Protection where its supporting requirements remain applicable;
- purpose limitation and data minimization for candidate, company, and recruitment data;
- explicit and recorded user consent where consent is the applicable processing basis;
- secure handling, retention, deletion, and access control for personal information and uploaded CVs;
- mechanisms for users to request access, correction, or deletion of their personal data, subject to applicable retention obligations; and
- recruiter and company verification with access to business documents restricted to authorized Administrators.

AI-generated recommendations shall remain advisory only and shall not replace human recruitment decisions.

6.15. Environmental and Platform Constraints

The system is designed as a modern web application using the following technology stack.

Layer	Technology
Frontend	Next.js
Language	TypeScript
Styling	Tailwind CSS
UI Components	Shadcn UI
State Management	Zustand
Drag & Drop	hello-pangea/dnd
Backend	Next.js API Routes
Database	PostgreSQL / MySQL
Authentication	JWT
AI Integration	OpenAI API or Another AI Model
Version Control	GitHub

The platform requires:

- Internet connectivity
- Modern web browser
- JavaScript enabled
- Email service for verification and notifications

6.16. Documentation Requirements

The project shall provide the following documentation:

- User Manual
- Recruiter User Guide
- Administrator Guide
- Installation Guide

- Deployment Guide
- API Documentation
- Database Schema Documentation
- System Architecture Documentation

Developer documentation shall be maintained alongside the source code.